

Agampreet Singh

Software Engineer · Backend Systems & Platform Architecture

— 8 YEARS

✉ agampreet_singh@yahoo.com ☎ +91 9041716953 🔗 linkedin.com/in/agampreetsingh 🐙 github.com/agampreetsingh
🌐 agampreetsingh.me

EXPERIENCE

JUL 2022 — PRESENT

DP World / Group SDE 2 — Tech Lead

Owner of the Quotations platform — a cross-functional pillar of DP World's global freight-forwarding business across **150+ locations**.

- Architecting **Scout**, an **AI-powered** product co-pilot for DP World logistics products — combining **LLM-powered** query resolution, contextual guided workflow tours, and pull-request based release summaries to replace manual onboarding walkthroughs and make product discovery self-serve.
- Designed Scout's **Gen AI application layer** — prompt templates, structured tool calling, retrieval pipeline, response guardrails, and traceable request flows — to keep LLM-powered features reliable, debuggable, and provider-agnostic.
- Built the **Quotations module from scratch** as part of DP World's **P100 programme** — live since Oct 2023; adoption scaled from **~20 to ~500 quotes/week (25x)**. Stack: Spring Boot · PostgreSQL · Hibernate · Redis · Azure EventHub · New Relic.
- Re-architected Quotes to be **config-driven** — zero-code support for new tenants, countries, and branches; decoupled CRM workflows to cut inter-service coupling — reducing tenant onboarding time from 3 Weeks to 3 hours
- Reduced Quote creation latency **15s → 2s** through targeted performance optimisation across the request path.
- Built a **high-performance PDF generator** from HTML templates — outperforms paid tools like Aspose; adopted across multiple DP World products.
- Tech-lead for the Quotations platform — set architecture direction, review bar, and delivery cadence for a team of **5+ engineers**; partnered with Product, Pricing, and Operations to land platform decisions across DPW divisions.
- Improved engineering quality, observability, and security posture — maintained **SonarQube coverage ≥ 80%** with **Maintainability rating A**, led **New Relic** migration for centralised observability, resolved VAPT findings, and standardised Java application templates including the **Notification Handler** now integrated by 8 internal apps.
- Recognised with the **Champions Award** (FY 2024–25) for delivery impact across the platform.

DEC 2020 — JUL 2022

Deutsche Telekom / Software Development Engineer

- Designed and developed an **authorization service** using Keycloak and Java, centralising access-control flows, admin dashboards, and security controls across internal applications.
- Built a dynamic **media file resizing** service using AWS Lambda + Python (Boto, MoviePy, PIL), enabling optimised image/video processing.
- Developed backend for a **rule-based chatbot** using tree architecture in Java Spring Boot.
- Designed backend APIs for a **template-based dynamic screen creator** for mobile apps, enabling configurable screen rendering without repeated app-side changes.
- Architected a reusable internal file-upload service; improved artifact management via **Nexus** jar-repository; improved Keycloak availability by externalising Infinispan cache.

FEB 2020 — DEC 2020

BlackRock / Analyst — Software Engineer

- Improved serialisation performance **10x faster** (15s → 1.5s) on the firm's Analytics platform.
- Contributed across the Analytics stack — Scala, Spark, Java, Spring Boot, Cassandra, MySQL — and gained hands-on experience resolving production incidents at scale.

JUN 2018 — JAN 2020

Nagarro / Associate Engineer

- Delivered solutions for a US-based ITM compliance system: implemented new user types and scaled email throughput **200 → 1000 / day** via GCP.
- Reverse engineered a legacy application and revamped the system from scratch, shipping the replacement into production.
- For a German airline client: upgraded codebase to **Swagger 2**, designed inter-service communication, and built new lounge-voucher types.

TECHNICAL SKILLS

BACKEND & CORE

Java · Spring Boot · Microservices · REST APIs · Hibernate · Maven

DATA & CACHING

PostgreSQL · MySQL · Redis · Apache Kafka · Cassandra

CLOUD & DEVOPS

Azure EventHub · AWS Lambda · GCP · Keycloak · New Relic · Grafana · SonarQube · Nexus

FRONTEND & LANGS

Node.js · Express · Angular · Vue.js · Python · Scala · Spark

GEN AI & AGENTS

LLM Workflows · Prompt Engineering · RAG · Tool Calling · Agentic Workflows · Guardrails

STRENGTHS

System Design · Performance Optimization · Team Leadership · Mentoring · Code Quality

EDUCATION

B.Tech, Computer Science & Engineering

Guru Nanak Dev University · 2014 — 2018

ENGINEERING FOCUS

Backend Platforms · Distributed Systems · Gen AI Applications ·
Agentic Workflows · Performance Optimization · Observability